College of Engineering – Department of Electronics Engineering

CURRICULUM

Bachelor of Science in Instrumentation and Control Engineering
Academic Year 2025-2026
Reference CMOs: CMO No. 86, S. 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, and CMO No. 40 s. 2021

Curriculum Description

The BSICE Program is dedicated to develop industry-ready engineers equipped with the technical expertise and practical experience required to meet the demands of modern industrial automation and process optimization. Anchored in an industry-based curriculum, the program emphasizes hands-on learning, real-world applications, and strong industry partnerships. Core principles of measurement, control, and automation are integrated with electrical, electronics, and mechanical engineering to ensure precision, efficiency, and safety in industrial operations.

Through internships, industry-based projects, and certification opportunities, students gain exposure to cutting-edge technologies and global standards, preparing them for leadership and innovation in key sectors such as manufacturing, energy, oil and gas, and smart technologies. The program also fosters ethical responsibility, teamwork, and continuous learning to ensure graduates remain competitive and adaptable in the rapidly evolving field of instrumentation and control engineering.

Program Education Objectives

The Instrumentation and Control Engineering alumni three to five years after graduation shall:
PEO 1: Practiced as specialist in solving complex instrumentation and control engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Assumed leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.
PEO 3: Engaged in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex instrumentation and control engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse

team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex instrumentation and control engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex instrumentation and control engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex instrumentation and control engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification/ Field / Course	No. of Hours/Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Sub-total	**14**	**0**	**14**
B. Natural/Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Sub-total	**6**	**6**	**8**

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C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Technopreneurship	3	0	3
Materials Science and Engineering	3	0	3
Environmental Science and Engineering	3	0	3
Computer Programming	0	3	1
Basic Occupational Safety and Health	3	0	3
Professional Orientation	2	0	2
Sub-total	**14**	**12**	**18**
D. Allied Courses			
Circuits 1	3	3	4
Circuits 2	3	3	4
Thermodynamics	3	0	3
Electronics Circuits: Devices and Analysis	3	3	4
Digital Principles and Microcontrollers	3	3	4
Control Systems Engineering	2	3	3
Manufacturing Automation	2	3	3
PLC Fundamentals and Programming	2	3	3
Sub-total	**21**	**21**	**28**
E. Professional Courses			
1. Core Courses			
Advanced Engineering Mathematics	3	0	3
Fundamentals of Instrumentation and Control Engineering	2	3	3
Fundamentals of Industrial Processes	2	0	2
Industrial Instrumentation	1	3	2

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Process Instrumentation and Control	2	3	3
Sensors Engineering	1	3	2
Process Operation and Modeling	0	3	1
Process Control and Automation	2	3	3
Analytical Instrumentation	3	0	3
ICE Capstone Design 1	0	3	1
ICE Capstone Design 2	0	3	1
Supply Chain Management for Instrumentation and Control Engineering	3	0	3
Reliability and Maintenance Engineering	3	0	3
Industrial Drives and Control	1	3	2
Sub-total	**23**	**27**	**32**
F. INDUSTRY PLACEMENT			
Industry Placement 1	640 hours		8
Industry Placement 2	640 hours		8
Sub-total	**1280 hours**		**16**
G. TECHNICAL ELECTIVE			
ICE Elective 1	1	3	2
ICE Elective 2	1	3	2
Sub-total	**2**	**6**	**4**
Total (Technical Courses)	**80**	**72**	**120**
II. NON-TECHNICAL COURSES			
A. General Education Course			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Sub-total	**24**	**0**	**24**

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B. Filipino/Literature/Mandated Courses			
Entrepreneurial Mind	3	0	3
Linear Algebra for Artificial Intelligence	3	0	3
Machine Learning and Neural Networks	2	3	3
Life and Works of Rizal	3	0	3
Sub-total	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 1	2	0	2
PATHFit 4 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 2)	2	0	2
Sub-total	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Sub-total	**6**	**0**	**6**
E. Institutional Mandated Course			
Foreign Language	**3**	**0**	**3**
Sub-total	**3**	**0**	**3**
Total (Non-Technical Courses)	**52**	**3**	**53**
GRAND TOTAL	**132**	**75**	**173**

SUMMARY	
Courses	**Number of Units**
I. Technical Courses	
A. Mathematics	14
B. Natural/Physical Sciences	8
C. Basic Engineering Sciences	18
D. Allied Courses	28
E. Professional Courses	32

F. Industry Placements	16
G. Technical Elective	4
II. Non-Technical Courses	
A. General Education Courses	24
B. Filipino/Literature/Mandated Courses	12
C. PATHFit 1-4	8
D. NSTP	6
E. Institutional Mandated Course	3
TOTAL	**173**

CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN INSTRUMENTATION AND CONTROL ENGINEERING

FIRST YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
MATH 101	Differential Calculus	4	4	0		
CHEM 101	Chemistry for Engineers	4	3	3		
ENGG 101	****Introduction to Engineering**	1	0	3		
ENGG 102	Engineering Drawing	1	0	3		
GEd 101	Understanding the Self	3	3	0		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 106	Purposive Communication	3	3	0		
PATHFit 1	Movement Competency Training	2	2	0		
NSTP 111	National Service Training Program 1	3	3	0		
	TOTAL	**24**	**21**	**9**		
			30			

Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Alangilan Campus
Golden Country Homes, Alangilan Batangas City, Batangas, Philippines 4200
Tel Nos.: (+63 43) 425-0139 local 2121 / 2221
E-mail Address: coe.alangilan@g.batstate-u.edu.ph | Website Address: http://www.batstate-u.edu.ph

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FIRST YEAR						
SECOND SEMESTER						
COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE-REQUISITE	CO-REQUISITE
			LEC	LAB		
MATH 102	Integral Calculus	4	4	0	MATH 101	
PHYS 101	Physics for Engineers	4	3	3	MATH 101	MATH 102
CpE 101	Computer Programming	1	0	3		
GEd 104	The Contemporary World	3	3	0		
GEd 108	Art Appreciation	3	3	0		
GEd 109	****Science, Technology and Society**	3	3	0		
GED 105	Readings in Philippine History	3	3	0		
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
	TOTAL	**26**	**24**	**6**		
			30			

FIRST YEAR						
MIDTERM SEMESTER						
COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE-REQUISITE	CO-REQUISITE
			LEC	LAB		
MATH 103	Differential Equations	3	3	0	MATH 102	
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101	
IMC 101	Foreign Language	3	3	0		
	TOTAL	**9**	**9**	**0**		
			9			

SECOND YEAR

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FIRST SEMESTER						
COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE-REQUISITE	CO-REQUISITE
			LEC	LAB		
ICE 202	Advanced Engineering Mathematics for ICE	3	3	0	MATH 103	
ME 101	Thermodynamics	3	3	0	PHYS 101, MATH 102	
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
ENGG 113	****Environmental Science and Engineering**	3	3	0	CHEM 101	
GEd 107	Ethics	3	3	0		
ICE 201	****Fundamentals of Instrumentation and Control Engineering**	3	2	3	2nd year Standing	
EE 207	Circuits 1	4	3	3	PHYS 101, MATH 102	
AI 101	Linear Algebra for Artificial Intelligence	3	3	0	MATH 102	
PATHFit 3	Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Traditional and Recreational Games)	2	2	0	PATHFit 2	
	TOTAL	**25**	**22**	**9**		
			31			
SECOND YEAR						
SECOND SEMESTER						
COURSE CODE	COURSE	UNITS	NO. OF HOURS		PRE-REQUISITE	CO-REQUISITE
			LEC	LAB		
ICE 203	****Fundamentals of Industrial Processes**	2	2	0	ICE 201	
ICE 204	****Industrial Instrumentation**	2	1	3	ICE 201	
ECE 409	Control Systems Engineering	3	2	3	ICE 202	
ENGG 114	Basic Occupational Safety and Health	3	3	0		
ECE 405	Electronics Circuits: Devices and Analysis	4	3	3	EE 207	

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EE 208	Circuits 2	4	3	3	EE 207	
ENGG 126	Professional Orientation	2	2	0	2nd year Standing	
AI 102	Probability and Statistics for AI	3	3	0	AI 101	
PATHFit 4	Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Team Sports, Basketball and Volleyball)	2	2	0	PATHFit 2	
	TOTAL	**25**	**21**	**12**		
			33			

THIRD YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
ENGG 126	Professional Placement 1	8	640hours		3rd yr standing, ENGG 125	
	TOTAL	**8**				
THIRD YEAR						
SECOND SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
ICE 301	****Process Instrumentation and Control**	3	2	3	ICE 204	
ICE 302	Sensors Engineering	2	1	3	3rd Year Standing	
ICEE 401	ICE Elective 1	2	1	3	3rd Year Standing	
MexE 206	****Manufacturing Automation**	3	2	3	ME 101	
MexE 305	PLC Fundamentals and Programming	3	2	3		MEXE 206, ECE 312

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ECE 312	Digital Principles and Microcontrollers	4	3	3	ECE 405	
ICE 404	ICE Capstone Design 1	1	0	3	ENGG 126	
	TOTAL	**18**	**11**	**21**		
			32			
THIRD YEAR						
MIDTERM SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
ICE 305	Industrial Drives and Control	2	1	3	EE 208, ICE 301	
ICE 403	****Process Control and Automation**	3	2	3	ICE 301	
GEdE 101	The Entrepreneurial Mind	3	3	0		
	TOTAL	**8**	**6**	**6**		
			12			

FOURTH YEAR						
FIRST SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
Ged 103	Life and works of Rizal	3	3	0		
ENGG 105	Technopreneurship	3	3	0	GEdE 101	
ICE 406	Supply Chain Management for Instrumentation and Control Engineering	3	3	0	4th year standing	
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CPE 101	
ICEE 402	ICE Elective 2	2	1	3	ICEE 401	
ICE 405	ICE Capstone Design 2	1	0	3	4th year standing	
ICE 401	Process Operation and Modeling	1	0	3	ICE 403	
ICE 304	****Reliability and Maintenance Engineering**	3	3	0	4th year standing	

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ICE 205	Analytical Instrumentation	3	3	0	ICE 204, CHEM 101	
	TOTAL	**22**	**18**	**12**		
			30			
FOURTH YEAR						
SECOND SEMESTER						
COURSE CODE	**COURSE**	**UNITS**	**NO. OF HOURS**		**PRE-REQUISITE**	**CO-REQUISITE**
			LEC	**LAB**		
ENGG 127	**Professional Placement 2**	**8**	**640 hours**		**Graduating**	
	TOTAL	**8**				
	CURRICULUM TOTAL	**173**	**132**	**75**		
		UNITS	LEC	LAB		

INDUSTRY BASED LEARNING

INDUSTRY -BASED CURRICULUM FOR BS INSTRUMENTATION AND CONTROL ENGINEERING			
YEAR	**Semester 1**	**Semester 2**	**Midterm**
1	Academic	Academic	Academic
2	Academic	Academic	
3	Industry Placement	Academic	Academic
4	Academic	Industry Placement	**GRADUATE**